Private pairs as a characteristic of a code, and why their number is not a lever for the gadget bounds on the Shannon capacity of C_7

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Abstract

Every lower bound on the Shannon capacity $\Theta(C_7)$ since July 2026 rests on gadgets in $C_7^{\boxtimes 5}$: a code, a family of *private pairs* (a vertex q outside the code with a single neighbour r in it), and an auxiliary independent set that must avoid a forbidden region and is rewarded for avoiding a penalised one. The bound is most sensitive to the number of private pairs. We study that number as a characteristic of a code and show why it is not, by itself, a lever. The two regions a pair creates are the intersection and the union of two closed neighbourhoods, so their sizes add up to $2 \cdot 3^d$: the shape of a pair moves cost between them but cannot lower it. For the Polak–Schrijver code, on which all the recent bounds are built, we prove that exactly eight private pairs exist and a ninth does not. For the codes with nine and ten pairs that we found, the forbidden regions are fixed by the geometry of the pairs: up to automorphism there are sixteen at nine pairs, the same for every such code, and against all of them the best auxiliary set we could build has 366 words, one short — exactly 3-opt optimal, and not proved maximum. We tabulate the five shapes of a pair with their costs and the candidate counts of every code we hold.

Keywords: Shannon capacity, strong product, independent set, odd cycle, private pair, gadget
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date	$\Theta(C_7) \geq$	from	witness
1971–2015	$343^{1/5}$, then $350^{1/5}$	[3, 4]	linear code; search
2019	$3.257865966783515950412 4\ldots$	[5]	$\alpha(C_7^{\boxtimes 5}) \geq 367$
Jul 2026	$3.2580207372932453595278\ldots$	[6]	$\alpha(C_7^{\boxtimes 10}) \geq 134753$
Jul 2026	$3.258789153908691016196765015520 6769\ldots$	[7]	gadget product
Jul 2026	$3.258805369885465572582975030623 8067\ldots$	[8]	stronger base gadget
Aug 2026	$3.258832620353266309121539051810 4754\ldots$	[9]	heterogeneous recursion

Table 1: Lower bounds on $\Theta(C_7)$. The last four entries start from the same 367-word code and the same eight private pairs, and differ in the recursion built on top.

1. Introduction

For a graph G the Shannon capacity is $\Theta(G) = \sup_d \alpha(G^{\boxtimes d})^{1/d}$ [1]. For the seven-cycle it is unknown. The Lovász bound gives $\Theta(C_7) \leq \vartheta(C_7) = 3.3176672073940953927\ldots$ [2], and the lower bound moved only in its fourth decimal place between 2019 and July 2026 (Table 1).

The structure behind the last four rows is Gao's *gadget* [7]: an independent set (the code), a family of *private pairs* — a vertex q outside the code with a single neighbour r in it — two transversals of their endpoints, and an auxiliary independent set. A product lemma multiplies gadgets, and the bound it yields is most sensitive to the number t of private pairs. Evaluated at hypothetical base profiles (a, t, s, o), the homogeneous recursion of [7, 8] gives

$$(367, 8, 367, 322) \mapsto 3.258805369885465572582975030623 8067\ldots,$$
$$(367, 9, 367, 321) \mapsto 3.259006200773852130 5802\ldots,$$

so one further private pair would be worth $1.736 \cdot 10^{-4}$ over the current record, while the record itself [9] exceeds what the homogeneous recursion extracts from the *same* base gadget by only $2.725 \cdot 10^{-5}$. A ninth pair would therefore improve the state of the art under any of the published recursions, by several times the spread between them.

This paper studies the number of private pairs as a characteristic of a code, and explains why that number is not, by itself, a lever. It does so in three steps, from the general to the particular. First (Section 3), the cost a pair imposes on the auxiliary set is conserved: the forbidden and the penalised regions a pair creates are the intersection and the union of two closed neighbourhoods, so their sizes add up to $2 \cdot 3^d$, and the shape of a pair can move cost between them but not lower it. Second (Section 5), for the

Polak–Schrijver code itself the count is at its stop: it admits exactly eight private pairs, and we prove that no ninth exists. Third (Sections 6–8), for the codes with nine and ten pairs that do exist, the forbidden regions are fixed by the geometry of the pairs rather than by the code, and against every one of them the best auxiliary set we could build is one word short.

The status of each claim is stated where it is made, and three kinds are kept apart: *proved* (by hand, with any finite check it rests on named); *exhaustive computation* (a complete enumeration of a finite object, reproducible from the published sets); and *search result* (the best found, which is not a bound). The results here were first deposited as two preprints [10, 11][1].

2. Gadgets, private pairs and the two regions

Vertices of $C_n^{\boxtimes d}$ are the elements of $\mathbb{Z}_n^d$; two distinct vertices are *confusable* iff their circular distance is at most 1 in every coordinate. $N(S)$ is the *closed* neighbourhood of a set S, as in [7, 9]; for $n \geq 3$ and a single vertex, $|N(v)| = 3^d$. A set is independent if no two of its elements are confusable.

Definition 1 (private pair [7])**.** Let I be independent. A pair (r, q) with $r \in I$, $q \notin I$ is a *private pair* for I if $N(q) \cap I = \{r\}$; r is its *centre* and q its *private neighbour*. The *candidates* of I are all such pairs.

Definition 2 (gadget [7])**.** A gadget consists of an independent set I of size a (the *code*); private pairs (r_i, q_i), $1 \leq i \leq t$, with pairwise disjoint endpoint sets; two complementary transversals $P^{\mathrm{H}}, P^{\mathrm{V}}$ of the endpoints — for every i exactly one of r_i, q_i lies in P^{H} and the other in P^{V} — both independent; and an independent auxiliary set X with $X \cap N(P^{\mathrm{H}}) \cap N(P^{\mathrm{V}}) = \emptyset$. With $O = X \setminus (N(P^{\mathrm{H}}) \cup N(P^{\mathrm{V}}))$, $H = X \cap N(P^{\mathrm{H}})$, $V = X \cap N(P^{\mathrm{V}})$, the profile is $(a, t, s, o, h, v) = (a, t, |X|, |O|, |H|, |V|)$.

We name the two sets in this definition:

$$F = N(P^{\mathrm{H}}) \cap N(P^{\mathrm{V}}) \qquad \text{(the forbidden region)},$$
$$U = N(P^{\mathrm{H}}) \cup N(P^{\mathrm{V}}) \qquad \text{(the penalised region)}.$$

[1] This paper merges and reorganises the preprints [10] (the theorem of Section 5 and the computations around it) and [11] (Sections 3, 4, 7 and 8).

X must avoid F, and every word of X in U is lost from o. Write $t^*(I)$ for the largest t for which a gadget with code I exists. The *shape* of a pair is $\delta = q - r \in \{0, \pm 1\}^d \setminus \{0\}$, and k is its number of non-zero coordinates.

The published base gadget has $I = I_0$, the 367-word set printed in the appendix of [5], and $t = 8$ with the eight pairs of [6]; its profile is $(367, 8, 367, 321, 26, 20)$ in [7] and $(367, 8, 367, 322, 26, 19)$ in [8, 9].

3. Cost is conserved

Lemma 3 (proved)**.** *Let $n \geq 3$ and let $P^{\mathrm{H}}, P^{\mathrm{V}}$ be any two sets of vertices of $C_n^{\boxtimes d}$. Then*

$$|F| + |U| \;=\; |N(P^{\mathrm{H}})| + |N(P^{\mathrm{V}})| \;\leq\; 3^d\,(|P^{\mathrm{H}}| + |P^{\mathrm{V}}|),$$

with equality iff the closed neighbourhoods of the vertices inside each of P^{H} and P^{V} are pairwise disjoint. In particular, for a single pair, $P^{\mathrm{H}} = \{r\}$ and $P^{\mathrm{V}} = \{q\}$,

$$|N(r) \cap N(q)| + |N(r) \cup N(q)| \;=\; 2 \cdot 3^d.$$

Proof. $|A \cap B| + |A \cup B| = |A| + |B|$ for finite sets, with $A = N(P^{\mathrm{H}})$, $B = N(P^{\mathrm{V}})$; and $|N(S)| \leq \sum_{s \in S} |N(s)| = 3^d |S|$, since for $n \geq 3$ the three values $a - 1, a, a + 1$ of every coordinate are distinct. □

The lemma is inclusion–exclusion and holds for any two vertices, private or not; what matters is its consequence for gadgets. A pair contributes $N(r) \cap N(q)$ to F and $N(r) \cup N(q)$ to U, and the two add up to $2 \cdot 3^d$, which is 486 in $C_7^{\boxtimes 5}$. *The shape of a pair redistributes its cost between the forbidden and the penalised region; it does not lower it.* A shape that makes F small makes U large by the same amount, and U is what o is paid from.

Lemma 4 (proved)**.** *For $n \geq 4$ and a pair of shape k, $|N(r) \cap N(q)| = 3^{d-k}\, 2^k$.*

Proof. Coordinatewise, $\{a - 1, a, a + 1\} \cap \{a + \delta - 1, a + \delta, a + \delta + 1\}$ has 3 elements if $\delta = 0$ and 2 if $\delta = \pm 1$, for $n \geq 4$; closed neighbourhoods in the strong product are products of the coordinate neighbourhoods. □

Both lemmas were also checked by direct count for every shape with $d = 3, 4, 5$ and $n = 3, 4, 5, 7, 9$ ($n = 3$ satisfies Lemma 3 and, as it must, not Lemma 4). For families the inequality is strict in practice: the nine-pair configurations of Section 7 have $|F| + |U|$ between 4050 and 4080, against $2 \cdot 9 \cdot 3^5 = 4374$.

k	shapes	$\lvert N(r) \cap N(q) \rvert$ (forbidden)	$\lvert N(r) \cup N(q) \rvert$ (penalised)	covered cells in box(q)	largest set found
1	10	162	324	16	366
2	40	108	378	8	367
3	80	72	414	4	366
4	80	48	438	2	366
5	32	32	454	1	366

Table 2: The five shapes of a private pair in $C_7^{\boxtimes 5}$, by the number k of non-zero coordinates of $q - r$. The first four columns are counted directly; the last is the largest independent set the exhaustive local construction of Section 4 produced carrying a private pair of that shape.

4. The five shapes

The 242 shapes in $C_7^{\boxtimes 5}$ fall into five orbits under the stabiliser of a vertex in $(D_7)^5 \rtimes S_5$, one per value of k (Table 2; exhaustive computation). The cost in F *falls* with k, so the two-coordinate shape is in the middle, not at the minimum. Every shape is realisable as a private pair — the expensive $k = 1$ included: all seven candidates of the Mathew–Östergård 350-word set [4] have $k = 1$.

The last column of Table 2 is an exhaustive computation over a finite list of inputs: for every word r of every code in Table 3 and of the swap plateau of Section 7, and every shape, put $q = r + \delta$, delete the other code words confusable with q, and add back an exact maximum independent set of the freed vertices that keep q private. In everything this produces, shapes other than $k = 2$ occur only one word below 367.

The column of covered cells explains why. A word w occupies the box $w + \{0, 1\}^5$ of 32 cells, and for $n \geq 4$ two vertices are non-confusable exactly when their boxes are disjoint; a private q meets the code's boxes only inside box(r), in 2^{5-k} cells. A pair of shape $k \geq 3$ therefore needs a box with at most 4 covered cells. In every 367-word code we can reach, every box outside the code has at least 5.

5. The Polak–Schrijver code has exactly eight private pairs

The definition of a gadget quantifies over transversals, which makes t^* look hard to bound. It is not: the transversal condition collapses to a condition on the private neighbours alone.

Lemma 5 (no mixing; proved)**.** *Let (r_i, q_i) and (r_j, q_j) be private pairs for I with $r_i \neq r_j$. Then r_i is confusable with neither r_j nor q_j.*

Proof. $r_i, r_j \in I$ and I is independent, so $r_i \not\sim r_j$. If $r_i \sim q_j$ then $r_i \in N(q_j) \cap I = \{r_j\}$, so $r_i = r_j$, contradicting the hypothesis. □

Corollary 6 (proved)**.** *Let (r_i, q_i), $1 \leq i \leq t$, be private pairs for I with pairwise distinct centres, and let $\sigma : \{1, \ldots, t\} \to \{\mathrm{H}, \mathrm{V}\}$ assign to each pair the transversal holding q_i. Then the endpoint sets are pairwise disjoint, and $P^{\mathrm{H}}, P^{\mathrm{V}}$ are both independent if and only if σ is a proper 2-colouring of the graph Γ on $\{q_1, \ldots, q_t\}$ whose edges are the confusable pairs $q_i \sim q_j$. Consequently $t^*(I)$ is the largest number of private pairs with distinct centres whose q-graph Γ is bipartite.*

Proof. Distinct centres give $r_i \neq r_j$; $q_i \neq q_j$ because $N(q_i) \cap I = \{r_i\}$ pins the centre; and $r_i \neq q_j$ because $r_i \in I$, $q_j \notin I$. So the endpoint sets are disjoint. Two r's are never confusable, and an r and a q of different pairs are never confusable by Lemma 5; r_i and q_i are confusable but lie in different transversals by construction. Hence a transversal can fail to be independent only by containing two confusable private neighbours, which is a monochromatic edge of Γ. □

So the only objects to examine are the candidates of I, a finite list that can be written down.

Proposition 7 (exhaustive computation)**.** *Let $I_0 \subseteq \mathbb{Z}_7^5$ be the 367-word Polak–Schrijver code as printed in [5]. Of the $16807 - 367 = 16440$ vertices outside I_0, exactly 8 have one neighbour in I_0, 254 have two, and 16178 have three or more; none has zero. The eight are exactly the private neighbours $q_0, \ldots, q_7$ of [6, 7], with the centres those papers give them.*

Proof. One pass over $\mathbb{Z}_7^5$, counting for each vertex outside I_0 its neighbours in I_0, with the integer predicate of Section 2. The empty zero-bucket says that I_0 is a maximal independent set, which is independently known [5]. The identification with the published pairs is a set comparison. □

Theorem 8 (proved)**.** *$t^*(I_0) = 8$. Moreover the admissible transversal assignments are exactly the proper 2-colourings of a graph with five connected components, 16 of them up to exchanging P^{H} and P^{V}, and one of them is the assignment used in [6, 7, 8, 9].*

Proof. By Corollary 6, $t^*(I_0)$ is computed from the candidate list of Proposition 7, which has eight entries with eight distinct centres; so $t^*(I_0) \le 8$. Its graph Γ has, in the numbering of [7], exactly the three edges

$$q_0 \sim q_1, \qquad q_2 \sim q_6, \qquad q_3 \sim q_5,$$

a matching on six of the eight vertices with two isolated vertices. A matching is bipartite, so all eight pairs may be used at once and $t^*(I_0) = 8$. Γ has $3 + 2 = 5$ components, hence 2^5 proper 2-colourings, $2^4 = 16$ up to a global swap; that the published assignment is one of them is checked directly. □

Corollary 9. *No gadget whose code is I_0 has $t \ge 9$. In particular the profile $(367, 9, 367, \cdot)$ is not reachable from the Polak–Schrijver code, under the homogeneous recursion of [7, 8] or any recursion that takes a gadget over this code as its base, including [9].*

Lemma 5 and Corollary 6 are proved by hand; the two finite facts they are applied to — the candidate list and the three edges — are computations over 16807 vertices that any reader can repeat from the printed set.

Two further questions around the theorem are settled by exhaustive computation. *All codes the Polak–Schrijver construction can produce:* step (v) of [5] extends a core of 327 words by a maximum independent set of an extension graph with 71 vertices and 85 edges, whose independence number is 40; that maximum is not unique, and enumerating all of them gives exactly 8 distinct codes of size 367, with candidate counts

$$5,\ 6,\ 6,\ 6,\ 7,\ 7,\ 7,\ 8,$$

the unique code attaining 8 being the printed one. *Codes of other sizes:* a gadget's code need not be maximum, but matching the bound of $(367, 8, 367, 321)$ through the published recursion would require 13 private pairs at $a = 366$, 16 at $a = 365$ and 23 at $a = 360$.

6. The count is a property of the code

Counting the candidates of every code we hold gives Table 3, a statistic that, as far as we can find, no paper reports. The counts do not follow size: the linear 343-word code [3] has none, the 350-word set of [4] has seven, the printed 367-word code has eight, and an independent rebuild of it, agreeing with the printed set in 365 of its words, has six. Among the 367-word codes

code	size	cand.	t^*	colourings	$\lvert F\rvert$ at t^*	by $k = 1, \ldots, 5$
linear code [3]	343	0	0	0	–	0, 0, 0, 0, 0
Mathew–Östergård [4]	350	7	7	64	1022–1118	7, 0, 0, 0, 0
auxiliary-set search, $t = 9$	363	20	18	96	2171–2308	2, 16, 2, 0, 0
repaired auxiliary set, $t = 10$	365	13	13	128	1582–1726	0, 13, 0, 0, 0
repaired auxiliary set, $t = 9$	366	12	12	64	1460–1561	0, 12, 0, 0, 0
pipeline code 4	367	5	5	16	552–568	0, 5, 0, 0, 0
I_0, our rebuild	367	6	6	16	686–708	0, 6, 0, 0, 0
pipeline code 0	367	6	6	16	686–708	0, 6, 0, 0, 0
pipeline code 5	367	6	6	16	686–708	0, 6, 0, 0, 0
pipeline code 6	367	6	6	16	688–708	0, 6, 0, 0, 0
pipeline code 1	367	7	7	16	820–844	0, 7, 0, 0, 0
pipeline code 2	367	7	7	16	822–848	0, 7, 0, 0, 0
pipeline code 7	367	7	7	16	822–848	0, 7, 0, 0, 0
pipeline, best	367	8	8	16	956–984	0, 8, 0, 0, 0
I_0, as printed [5]	367	8	8	16	956–984	0, 8, 0, 0, 0
pipeline code 3	367	8	8	16	956–984	0, 8, 0, 0, 0
X of [7], as a code	367	9	9	16	1098–1128	0, 9, 0, 0, 0
ten-candidate code	367	10	10	16	1240–1268	0, 10, 0, 0, 0
$o = 322$ auxiliary set	367	10	10	16	1240–1268	0, 10, 0, 0, 0

Table 3: The candidate count of every code we hold in $C_7^{\boxtimes 5}$. Candidates are the vertices with exactly one neighbour in the code; t^* is the largest usable family (Corollary 6); colourings are counted up to exchanging P^{H} and P^{V}; $\lvert F\rvert$ ranges over them. Exhaustive for each file.

the counts run from 5 to 10, every candidate has shape $k = 2$, and for a fixed code the forbidden region varies over the admissible colourings by at most 3.21%. The codes with nine and ten candidates are new: one of them is the auxiliary set of the published gadget [7], read as a code.

7. Sixteen forbidden regions

A private pair is also a move: $I \mapsto I - r + q$ keeps size and independence; it is the exchange used in [9]. Closing every 367-word code we hold under these moves, and under exhaustive exchanges of two and three words, gives a finite plateau that falls into 51 classes under $\mathrm{Aut}(C_7^{\boxtimes 5})$ (exhaustive computation; exchanges of four words were time-limited and found nothing new). Each connected component is a product of five independent three-state switches, $3^5 = 243$ codes; the five switches sit on the five cyclically adjacent pairs of coordinates and are permuted by an automorphism of order 5 that cycles the

coordinates. The number of components depends on the starting codes and is not structural.

Two classes have nine candidates, and one class has ten. An automorphism g moves the forbidden region to $g \cdot F$ and leaves $\alpha(C_7^{\boxtimes 5} \setminus F)$ unchanged, so the variables are the code up to Aut and the colouring; sweeping automorphisms of the code adds nothing. All 208 configurations with $t \geq 9$ give, up to automorphism, exactly 16 forbidden regions at $t = 9$ and 4 at $t = 10$, with

$$1098 \leq |F| \leq 1128 \ (t = 9), \qquad 1240 \leq |F| \leq 1268 \ (t = 10),$$

and *each of the sixteen is produced by every nine-pair class* (exhaustive computation). The forbidden region is a function of the geometry of the nine pairs, which are the same switches in every code, and not of the rest of the code. Of a nine-pair region, $972 = 9 \cdot 108$ vertices come from the pairs themselves and 126 to 156 from cross terms between pairs.

8. One word short

At $t = 8$.. With $t = 8$ forced on I_0, the remaining parameter is X. The candidates for X are the eight pipeline codes and their images under $(D_7)^5 \rtimes S_5$, a group of $14^5 \cdot 120 = 64{,}538{,}880$ elements; with every admissible colouring the sweep covers 8,260,976,640 triples (source, colouring, automorphism), exhaustively, since for fixed permutation and reflections all translations are scored at once by a cyclic cross-correlation. The forbidden intersection is never empty:

words of X in F	0	1	2	3	4	5	6	7
triples	0	4	33	101	209	358	965	6,060

This explains a step of the published construction stated there without comment: X is the image $T(I_0)$ under $T(w) = (2 - w_1, w_3, w_0, 2 - w_2, w_4)$ with the single word $(2, 4, 6, 3, 5)$ replaced by $(1, 5, 6, 3, 5)$ [7, 9]. The replacement is unavoidable, and $(2, 4, 6, 3, 5)$ is exactly the one word of $T(I_0)$ in F. Exact repair of the near misses reaches $s = 367$, $o = 322$, the profile of [8], recovered from another direction and not improved. A 367-word admissible auxiliary set was found under exactly one of the 16 colourings, the one the literature uses; for the other fifteen this is a search result (1500 s of fixed-cardinality search each), not a proof.

At $t = 9$ and $t = 10$.. Against each of the 20 regions of Section 7, every image under $(D_7)^5 \rtimes S_5$ of one representative of each of the 51 classes was scored exhaustively, 65,829,657,600 images in all. No image avoids any region; the fewest words of an image in F is 1. Every image within three words of admissible was repaired exactly, and the best results are

$$366 \text{ words at } t = 9, \qquad 365 \text{ at } t = 10.$$

All 108 repaired 366-word sets are exactly 3-opt optimal inside $G_F = C_7^{\boxtimes 5} \setminus F$: no window of at most three words can be traded for one more, decided by counting blockers. Only 2 or 3 allowed vertices are blocked by a single word of such a set.

Status. This is a computational result, not a proof that 367 words do not fit outside F. The cheap upper bounds cannot see F: the Lovász bound gives 401 whether F is empty or removes 1268 vertices, because it bounds the whole graph, and a box-volume count is weaker still. An honest bound needs ϑ of the induced subgraph on 15,679 to 15,709 vertices, and F breaks the symmetry that would reduce it.

Why a cheaper shape does not obviously help.. With $|U|$ taken without overlaps, the fraction of its random share that X may touch at $t = 9$ and 367 words is 0.879, 0.754, 0.688, 0.651, 0.628 for $k = 1, \ldots, 5$; the best auxiliary sets on record reach 0.7678. By Lemma 3 the shapes that would shrink F are the ones that ask most of o, and at 366 words no shape allows more than 0.68 of the random share at $t = 9$, 13 or 16. These are distances to the best known search, not impossibility results.

9. What remains open

On I_0 there is no ninth pair (Theorem 8). On the codes with nine and ten pairs that we hold, the best auxiliary set falls one word short at nine pairs and two at ten, against forbidden regions fixed by the geometry of the pairs; that is a statement about the codes we hold, not about all codes. Two things would change it.

1. *A 367-word code with a deeper hole*: a box covered in at most 4 cells, which a pair of shape $k \geq 3$ needs. None of the codes we can reach has one; finding one requires 367-word codes from outside the known family, which is the open problem around $\alpha(C_7^{\boxtimes 5})$ itself, where even 368 is open.

2. *An upper bound that sees F*, of ϑ strength on the induced subgraph, which would turn the one-word gap of Section 8 into a theorem or refute it.

10. Methods and reproducibility

All computations are exact integer computations. Independence is checked by a standalone C verifier: for $n \geq 4$ two distinct vertices of $C_n^{\boxtimes d}$ are adjacent exactly when their boxes meet, so a set is independent iff its $|I| \cdot 2^d$ box cells are all distinct; for a 367-word set that is 11,744 cells. The verifier is held to cases where it can fail: 1177 single-vertex corruptions against independently computed ground truth, and a deliberately defective build that must be rejected. Each further estimator was calibrated in the same way before use: the new group sweep reproduces the $t = 8$ histogram above exactly, the repair reaches 367 at $t = 8$, the window search recovers 367 from deliberately damaged sets, and the automorphism checker rejects a non-automorphism. Predictions were registered before each computation.

The software was written, and the computations run, through Claude Code (Anthropic) under the author's direction, following preregistered stage plans; every number in this paper is reconciled by script against the stored results, an unsourced number counting as a failed build. Theorem 8 and the lemmas do not rest on the software: their proofs are by hand, plus finite checks that can be repeated from the printed set [5].

Data availability.. The sets, the verifier, the scripts that produce every number here, the preregistrations and the machine reconciliation of the numbers are archived with the two preprints this paper is assembled from, doi:10.5281/zenodo.22979509 and doi:10.5281/zenodo.23234251 [10, 11, 12].

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.. During the preparation of this work the author used Claude and Claude Code (Anthropic) in order to draft and revise the text. After using this tool/service the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The use of the same tools in the research itself is described in Section 10.

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